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Studierichting: Informatica
Oefeningenreeks 2E nr. 18.6

$$4 \cdot \log_x \sqrt{3} + \log_x 3^{\frac{1}{3}} + \log_x \frac{1}{9} + \frac{1}{3} = 0$$
$$\Leftrightarrow \log_x \sqrt{3}^4 + \log_x 3^{\frac{1}{3}} + \log_x \frac{1}{9} = -\frac{1}{3}$$
$$\Leftrightarrow \log_x 9 + \log_x 3^{\frac{1}{3}} + \log_x \frac{1}{9} = -\frac{1}{3}$$
$$\Leftrightarrow \log_x \left(9 \cdot \frac{1}{9}\right) + \log_x 3^{\frac{1}{3}} = -\frac{1}{3}$$
$$\Leftrightarrow \log_x 3^{\frac{1}{3}} = -\frac{1}{3}$$
$$\Leftrightarrow x^{-\frac{1}{3}} = 3^{\frac{1}{3}}$$
$$\Leftrightarrow x^{-\frac{1}{3}} = \left(\frac{1}{3}\right)^{-\frac{1}{3}}$$
$$\Leftrightarrow x = \frac{1}{3}$$

References